MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION (COMBINED TECHNICAL) FOR RECRUITMENT TO THE POSTS OF AGRICULTURE & ALLIED SERVICES, GOVERNMENT OF MIZORAM.

(NOVEMBER-2025)

HORTICULTURE SCIENCE PAPER-II

Time Allowed : 3 hours FM : 200

SECTION - A *(Multiple Choice questions) (100 Marks)*

All questions carry equal mark of 2 each. Attempt all questions.

*This Section should be answered only on the **OMR Response Sheet** provided.*

Notes regarding Negative Marking :

(i) There will be penalty (negative marking) for every wrong answer given by a candidate in the MCQ type questions.

(ii) For each question for which a wrong answer has been given by the candidate, one-third of the marks assigned to that question will be deducted as penalty

(iii) If a candidate gives more than one answer to one question, it will be treated as a wrong answer even if any one of the given answers happens to be correct and there will be same penalty as above for that question.

(iv) If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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1. Parthenocarpy is a characteristic feature of:

(a) Banana (b) Guava

(c) Litchi (d) Citrus

2. Informal style of gardening is found in:

(a) Mughal gardens (b) Italian Gardens

(c) French gardens (d) Japanese gardens

3. Fruit type of Litchi is:

(a) Berry (b) Single Seeded Nut

(c) None (d) Etaerio of achens

4. Kinnow is a cross between:

(a) Willow leaf × King (b) King × Willow leaf

(c) Mandarin × grapefruit (d) Mandarin × pummelo

5. Which is the non climacteric fruit?

(a) Orange (b) Mango

(c) Banana (d) Fig

6. Little leaf mango is due to:

(a) Zn deficiency (b) Iron deficiency

(c) Calcium deficiency (d) Mo Toxicity

7. Polyembyony is seen in:

(a) Ber (b) Jamun and Citrus

(c) Grape (d) Guava

8. Strawberry is octaploid having 2n=8X=

(a) 24 (b) 34

(c) 56 (d) 26

9. Metaxia is commonly observed:

(a) Papaya (b) Date

(c) Avocado (d) Aonla

10. Bunchy top of banana is caused by:

(a) Fungus (b) Bacteria

(c) Virus (d) physiological factor

11. Tomato leaf curl virus is transmitted by:

(a) Thrips (b) White fly

(c) Aphids (d) fruit fly

12. Arka Abhed is a hybrid variety of:

(a) Brinjal (b) Tomato

(c) French bean (d) Cow pea

13. Which potato variety is known for early maturity?

(a) Kufri Jyoti (b) Kufri Karan

(c) Kufri Chandramukhi (d) Kufri Frysona

14. Chromosome number of French bean is:

(a) 18 (b) 45

(c) 10 (d) 22

15. Which among the following fertilizers is least suitable for use in fertigation?

(a) Nitrogen fertilizer (b) Phosphorus fertilizers

(c) Potassium fertilizers

16. Arka Nikita is the variety of:

(a) Brinjal (b) Onion

(c) Okra (d) Tomato

17. Buttoning is a physiological disorder in:

(a) Garlic (b) Onion

(c) Radish (d) Cauliflower

18. Arka Khyati is avariety of:

(a) Onion (b) Chilli

(c) Tomato (d) Okra

19. Long day type of Garlic is:

(a) Agrifound Parvati (b) Yamuna Safed

(c) G 282 (d) Agrifound white

20. Arka Veera is the variety of:

(a) Brinjal (b) Tomato

(c) Cucumber (d) Cow pea

21. Baradari is an important feature of which garden?

(a) Mughal (b) French

(c) Chinese (d) English

22. A popular Japanese arrangement of flowers is known as:

(a) Ikebana (b) Tatabana

(c) Suchigama (d) A bouquet

23. Heterostily is a common feature in:

(a) Petunia (b) Primula

(c) Gerbera (d) Ageratum

24. Gladiolus is propagated by:

(a) Corm (b) Bulb

(c) Rhizome (d) Tuber

25. Bent neck is a common physiological disorder of:

(a) Rose (b) Anthurium

(c) Chrysanthemum (d) Bougainvillea

26. Which of the following is foliage annual?

(a) Aster (b) Cosmos

(c) salvia (d) Kochia

27. Arka Archana is a variety of:

(a) Hibiscus (b) Aster

(c) Lily (d) Tulip

28. Fairy rings of lawn is due to:

(a) Fungus (b) Bacteria

(c) Insect (d) Nematode

29. Largest in producing and trading of cut flower in the world is:

(a) USA (b) Israel

(c) India (d) The Netherlands

30. Which of the following greenhouse glazing materials may be considered most suitable for long term use?

(a) Polyvinyl chloride film
(b) Acrylic
(c) Glass
(d) Polycarbonate

31. Among honey bees, workers are:

(a) Male
(b) Female
(c) Hermaphrodite
(d) Male and Female both

32. Upto how many eggs can a queen bee lay in one day?

(a) 500
(b) 1000
(c) 1500
(d) 2000

33. The male bees in a honey bee colony are known as:

(a) Workers
(b) Drones
(c) Queen
(d) Nymphs

34. Which of the following bee is developed from unfertilized eggs?

(a) Worker
(b) Drones
(c) Queen
(d) Both the worker and queen

35. Which of the following is contained in medicinal yam?

(a) Digoxin
(b) Cravone
(c) Diosgenin
(d) Solasodine

36. Methyl chavicol is present in significant amount in which of the following aromatic plants?

(a) Lemongrass
(b) Jasmine
(c) Basil
(d) Celery

37. Cinchona bark contains which important compound?

(a) quinine
(b) nicotine
(c) ephedrine
(d) cocaine

38. Recommended spacing for dwarf variety of papaya 'Pusa Nanha' in high density planting is:

(a) 1.25m × 1.25m
(b) 1.50m × 1.50m
(c) 1.75m × 1.75m
(d) 2.0 m × 2.0 m

39. What is the scientific name of Shiitake mushroom?

(a) Pleurotus florida
(b) Lentinula edodes
(c) Colocybe indica
(d) Auricoloria polytricha

40. Which substrate is commonly used for oyster mushroom cultivation?

(a) Compost
(b) Sand
(c) Straw
(d) Clay

41. Cherelle wilt is a physiological disorder affecting:

(a) Coconut
(b) Cashewnut
(c) Cocoa
(d) Coffee

42. Budded Rubber plants ideal height for tapping(from the bud union) is:

(a) 50 cm girth and 125-150 cm height
(b) 35 cm girth and 90-100 cm height
(c) 90 cm girth and 180-250 cm height
(d) 75 cm girth and 160-180 cm height

43. State known to be as Spice Garden:

(a) Tamil Nadu
(b) Kerala
(c) Assam
(d) Andhra Pradesh

44. The causal organism of leaf rust in coffee is:

(a) Exobasidium vexans
(b) Hemileia vestatrix
(c) Fomes noxius
(d) Ganoderma lucidum

45. What is the primary goal of protected cultivation?

(a) To enhance the visual appeal of agricultural fields
(b) To control the microclimate surrounding plants for optimal growth
(c) To minimize the use of agricultural machinery
(d) To increase the dependency on natural weather

46. Picrocrocin is present in:

(a) Saffron
(b) Sweet Flag
(c) Dill
(d) Clove

47. Linalool is important chemical content of:

(a) Cumin
(b) Fennel
(c) Coriander
(d) Clove

48. Acorus calamus is the botanical name of :

(a) Dill
(b) Aniseed
(c) Fenugreek
(d) Sweet flag

49. Plant part used in saffron is:

(a) Fruits
(b) Seeds
(c) Stigma
(d) leaves

50. Commercial part of clove is:

(a) Fruit
(b) Seed
(c) Unopened flower bud
(d) Leaves

SECTION - B *(Conventional Type) (100 Marks)*

*This Section should be answered only on the **Answer Sheet** provided.*
Marks for each question is indicated against it.
*Attempt **any 5 (five)** questions.*

1. Describe the cultivation and production technology of grape under the following heads. **(5×4=20)**

(a) Climate and soil
(b) Varieties
(c) Training and pruning
(d) Physiological disorder
(e) Pest and Diseases management

2. Discuss the cultivation of Ginger under the following heads. **(5×4=20)**

(a) Propagation
(b) Varieties
(c) Major pest and their management
(d) Major diseases and their management
(e) Post harvest management

3. Write notes on *any four* of the following: **(4×5=20)**

(a) Flowers and sex expression in Papaya.
(b) Planting, Training and Pruning in Kiwi.
(c) Planting, Training and Propagation in Strawberry.
(d) Explain varieties, use of growth regulators and planting methods in Pineapple.
(e) Irregular bearing, major pests and Physiological disorder in Mango.

4. Write notes on *any four* for the following: **(4×5=20)**

(a) Physiological disorder, pest and diseases management in Tomato.
(b) Role of plant growth regulator in vegetable crops.
(c) Define bolting. What are the causes and means of control of bolting in onion?
(d) Discuss cultivation of cabbage under the following heads : soil and climate, seed rate and spacing, manures and fertilizers.
(e) Cultivation of okra on the following points: seed rate, sowing time, spacing, harvesting and yield.

5. Answer the following questions in detail: **(4×5=20)**

(a) Propagation, harvesting and post harvest management in rubber.
(b) Post harvest management and processing of tea.
(c) What are important elements of landscaping? Give main features of Mughal garden, Japanese garden and English garden.
(d) Use of greenhouse technology in boosting crop productivity.

6. Answer the following questions in detail: **(4×5=20)**

(a) What do you understand by maturity indices? Give maturity indices for musk melon, water melon, onion and garlic.

(b) Define unfruitfulness. Differentiate between gametophytic and sporophytic self-incompatibility along with examples. Describe Monoecious, Dioecious and Hermaphrodite fruit plants.

(c) Cultivation of Broccoli on the following points: seed rate, sowing time, spacing, harvesting and yield.

(d) Discuss the origin, cultivation practices and pests and diseases of potato.

7. Provide detailed answers to the following questions: **(4×5=20)**

(a) What is bonsai? How is watering done in Bonsai? Write five plants suitable for making Bonsai. What is the ideal time of planting Bonsai?

(b) Discuss the harvesting, grading, packaging and storage techniques followed in anthurium.

(c) How can dormancy be checked in Gladiolus? What are important diseases in Gladiolus?

(d) How are Carnation propagated? How is disbudding done in carnation and what is the optimum stage of harvesting in carnation?

8. Answer the following questions: **(4×5=20)**

(a) Sex expression and sex ratio in Cucurbits.

(b) Discuss the role of growth regulators in fruit set, ripening and sex expression in vegetable crops.

(c) Discuss cultivation of cabbage under the following heads : soil and climate, seed rate and spacing, manures and fertilizers.

(d) Protected Cultivation: importance, scope and status.

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